

Quiz – Knowledge & Inference

Instructions: This is a multiple-choice question section. For each question, choose **only one** correct answer.

* **Enter your name and ID**

* (1)Name

请输入

* (2)Student ID

请输入

1. [单选] **FOL** (0.5分)

What does the following FOL statement mean?

$\forall x (\text{Student}(x) \rightarrow \exists y (\text{Course}(y) \wedge \text{Takes}(x, y) \wedge \forall z (\text{Takes}(z, y) \rightarrow z = x)))$

- ☐ There is a course that every student takes, and no other student takes it.
- ☒ Every student takes at least one course that is taken by only that student
- ☐ Every student takes exactly one course, and no two students share a course.
- ☐ For every course, there is a student who takes it and that student takes no other course.

2. [单选] **Definite Clause** (0.5分)

Which of the following is NOT a definite clause?

- ☐ Option 1

$\text{Parent}(x, y) \wedge \text{Female}(x) \rightarrow \text{Mother}(x, y)$

☐ Option 2

Ancestor(x, y)

☒ Option 3

Likes(x, y) $\vee$ Hates(x, y) $\leftarrow$ Person(x) $\wedge$ Person(y)

☐ Option 4

$\rightarrow$ True

3. [单选] Propositional Inference (0.5分)

Given the definite clause knowledge base below, and observations {H, G}, the query is D. What is NOT a possible order of atoms derived by forward chaining?

$A \rightarrow B$
 $B \wedge C \rightarrow D$
 $E \rightarrow C$
 $F \rightarrow A$
 $G \rightarrow E$
 $H \rightarrow F$

☐ F, A, B, E, C, D

☐ E, F, A, B, C, D

☒ E, F, B, A, C, D

☐ F, E, A, C, B, D

4. [单选] Reasoning with uncertainty (0.5分)

In Wumpus World, each cell independently contains a pit with probability 25%. The agent is safe at cell (1,1). The breeze sensor of the agent is probabilistic: (1) if at least one adjacent cell has pit, it returns Breeze with 90% chance; (2) if no adjacent cell has pit, it returns Breeze with 10% chance.

Now, the agent senses breeze at (1,1). What is the probability that (1, 2) has a pit?

☒ 50%

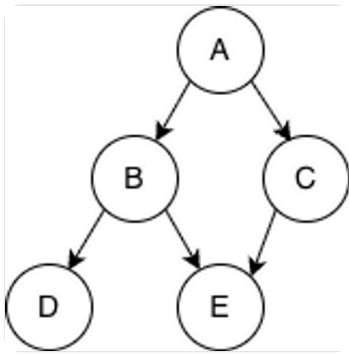
☐ 40%

☐ 30%

☐ 20%

5. [单选] **Bayesian Network** (0.5分)

Given the Bayesian Network below, which of the following is true:



- ☐ A and E are independent
- ☒ D and E are independent given B
- ☐ C and E are independent given A
- ☐ A and E are independent given B

提交